

I: So, just to start here, a note for the research with covid 19 disruption. We just wanted to acknowledge that changes teaching and learning. So for any of these questions that I'm going to ask you can either answer with reference to your normal teaching pre covid or your teaching practices now that you've adapted or both whatever you feel like.

I: All right. So, first we just want to start getting some background. So could you briefly describe your experience teaching undergraduates, including how your teaching relates to your current or past research, and in which of the courses you teach do your students work with data?

UCSB03: So I've taught two different undergraduate courses here. The first one is [course redacted]. And I'm not teaching that this year but I have a lot in the past. And this is basically the application of statistics in [discipline redacted], so it's entirely about learning methods to analyze data, which is what I do in large parts of my research is analyze data. And then this fall, I'm teaching a brand new course remotely, asynchronously, with great difficulty, which is [data wrangling for [discipline redacted]]. And this, this one is specifically about handling data. More handling than actually analyzing though we do a little bit of analyzing. It's trying to teach students skills for how to take the kind of dirty data you get in the real world and massage it into something that's usable for analysis or for making decisions, that kind of thing.

I: Great, I guess we'll have to talk to you more on that because that's also the kind of workshops we try to put on the Collaboratory So yeah, that's good to hear.

UCSB03: Alright.

I: So, as far as those two courses go, the next question is how do your students usually obtain the data they engage with? So I'm going to ask questions on how they get data. You know, then how you work with it in the course and then what kind of training and support there is. But first we're wondering how they get that data. Do they collect and generate it themselves, do they search and identify pre existing data, or do you provide the data sets for them?

UCSB03: We provide the data sets for them.

I: You provide them. Okay. Can you describe the general process that students go through then?

UCSB03: In the introduction to [class redacted], we teach statistical or [disciplinary approach redacted] techniques. And we then give them some mild instruction on using an [disciplinary approach redacted] software package. We supply them data and then we ask them to do a particular kind of analysis, basically. So the idea is that in lecture they've learned some theory they've seen some examples. And then they're being asked to apply what they learn. In the new course that we're doing the lectures are all asynchronous.

And what we do is we supply them with data and ask them to do a specific analysis which requires them to write some computer code, and then requires them to write up the results that they get.

I: Okay. And then, have you found in either of those courses that they're may be different challenges or the same challenges, that students have challenges with?

UCSB03: Yeah. Very, this is going to be, may sound a little strange, but students seem to be decreasingly computer savvy. And I think some part of this is the shift to everybody being on cell phones all the time. But I have had, I'll give you an example from an earlier course, I had a student come to my office and say okay I've got to turn in this write up. And I got this output on the screen and the [discipline redacted] package. How do I get this stuff into my Microsoft Word file? And I said, copy and paste. And he said, oh, how do you do that? And literally he was so thrilled when I could show him how to copy and paste. And I'm not the only one who's had this experience. So the course I'm teaching now where the delivery mechanism is R, and we are assuming that students have never coded anything before. There are some of the students who are in total, complete bewilderment as to how to think about writing a piece of computer code, and you know we give them lots of examples and we're trying to do hand holding and. But for some of them, it's just a complete mystery, and some part of this is in both courses but more, particularly in this new one. It's not about memorization. It's not. I mean, you know, we give the lectures, but it's not about regurgitating what's in the lecture, it's being actually asked to go out and do things. And I think in a lot of courses here at UCSB, partially because class sizes are so large, that's actually rather unusual. And for some of them, I think they're having some trouble learning how to deal with this.

I: Yeah, and I know it's it's definitely a different mind space that you have to be in like, you know, coding and programming requires a different kind of logic than you have to use, and other you know other circumstances under that was it took me a little bit to get my mind around that when I started.

UCSB03: Yeah, I mean, that's really true, but I think I'm saying there was actually something a little bit more to it. In a lot of courses, you're shown some material, and you're then asked to be able to feed that material back on an exam, maybe in you know a slightly new problem you haven't seen before.

Whereas, what we're doing is we're saying look, here's the task you'd like to accomplish. Here's an example of that kind of task and how it could be done. Now here is a new task. You now have to figure out which of the tools you've learned to apply and so forth. And for some people that is just fine but for a lot of them. This idea that they have not been told what to do, is actually pretty hard.

I: I see, I see what you mean there. Yeah, and I know I TA'd for some courses in grad school and we did a lot of HTML coding. So, you know, I think more people now are familiar with that, than like R coding. But even then, I mean, it was unzipping files that people had difficulties with.

UCSB03: Well right so we're having students, we're very fortunate that Google donated computer time and so we have a server running in a Google Cloud, and students write code on that. And then they have to push a button marked export to download their code to their own computer so they can turn it in and things, a bunch of them have no idea how to do that. And having had it explained they still don't really know how to do it. Well, you know, if you've never used anything like this on a computer before, then it's, you know, I wouldn't want to criticize somebody for not easily knowing how to do that. But it just seems very strange. So I have a colleague in the [department redacted] who tells me she's had students write term papers on their cell phone.

I: Oh yeah, I've heard that too. I was reading a piece a couple weeks ago that was kind of talking about this problem with students that are coming in that are not as computer savvy, and they're

talking about how it has a lot to do with how you know UI design has gotten so good that you don't really need to do much yourself anymore in these apps.

UCSB03: Yeah, I think that's right. And this is something which has sort of surprised me because I just assumed that over time, more and more people would be computer savvy, and it does not seem to be going that way.

I: Yeah, that's interesting. I think I was caught right in the middle of that transition there. But getting back to, you know, the data sets that you're working with. How do you go about obtaining those data sets that you use for teaching and vetting them? And then do you find any challenges in finding data sets for teaching?

UCSB03: Yes, so we do find some challenges. So I try very hard to use real data, rather than toy examples. And because it's their [discipline redacted] classes we typically use [discipline redacted] data or business data but mostly [discipline redacted]. The particular challenge that makes stuff hard to find is it would really be nice to get data from businesses. And they are almost never willing to do that because you know I don't know exactly what they're worried about but there's nothing in it for them, I guess.

I: Yeah, that's interesting. I tuned into the very end there's like the, I think it's a new center but Center for Responsible machine learning, on campus and they did a little conference last week, and a little bit I was tuning in on there talking about how they were actually able to track Covid cases better because they had gotten data from insurance companies. I was just sitting there wondering how they managed to convince the insurance companies to give them that data.

UCSB03: Right. I should say that this has been a big change in [discipline redacted] research, more and more people have been doing things to get access, sometimes confidential from industry sometimes from governments. But when you're doing it, I mean so sometimes people will invest a couple of years trying to get data. You can't do that for teaching. Right. And so, lots of data is, you can get access to it but only on a confidential basis well that doesn't do any good for teaching. So one of the best datasets, I actually, I happen to have is, you know, [name redacted].

I: [clarifying question]?

UCSB03: No, you know, [name redacted]. He runs the, I don't remember the name of it but institutional research. Great guy is actually a [PhD graduate] originally I think I saw years ago. He basically gave me data on all the student grades for a year. And now, of course he very carefully took out the names which we care about, and put in fake identifying numbers and we, you know, did some other things to protect privacy, so it was perfectly safe. But this lets me, in teaching, use actual grades and talk about how different departments grade and things like that. Well, you know, that's really good for student motivation, something they can really identify with. And so that's an example of something that's really good but it's hard to get that sort of stuff.

I: Yeah, I know we've been in for library workshops aimed at the librarians we've been trying to use more of our own data too but we have to do that as well because if it's like the door counts or, who's swiping cards, you know, we all have to do de-identify all of that before using it. Yeah. Okay, well, I'm gonna move on to the next question, which is more on working with data. So generally, how do

the students manipulate, analyze, or interpret the data in your course? You know what kind of tools or software do they use? Do you expect them to come in with prior knowledge, things like that?

UCSB03: Okay, so in the [discipline redacted] that I'm not teaching at the moment, we use Eviews, which is an [discipline redacted] software package. We certainly do not expect them to have any background in using it whatsoever. In the class that I'm teaching now it's been done in R. And we not only don't expect them to have any background in R, we sort of assume they've never programmed anything at all. We, to some extent, kind of expect students may have used Excel at some point. We don't, I don't use that very heavily. And it turns out there are quite a few students who are not very familiar with Excel which I find a little shocking but there you have it.

I: Yeah. And with those courses are they typically freshmen?

UCSB03: No, no, no, no, they're their upper division courses, the [discipline redacted] is a required upper division class. And this new class I'm teaching is an [course level redacted].

I: Okay, so that's pretty surprising that some of them haven't used Excel much. And then, to what extent are each of those tools do you feel they're important for teaching? I mean, it sounds like the [discipline redacted] and R are kind of the core of the courses but yeah.

UCSB03: So, so, right. So, using Eviews, which is widely used all around the country for the thing. Some of the other programs which people might substitute for STATA. I use Eviews because I think it's better, I'm familiar with it. And then when for this course I'm doing now we chose R. Mostly because it's very widely used in data science, and some of our accounting firms that help support the department had expressed some interest in students, knowing more, knowing some R. And actually, the accounting firms donated some, one in particular donated, some money to help develop the course, which was very important. Yeah, when you teach a course which is very heavily tech. Well, there's a lot of tech involved in getting the damn thing running, that's especially true doing it remotely. So we had a crew of three graduate students, and two undergraduates working all summer, putting together material, figuring out the technology and all those sort of things like that, and it was invaluable, we would not have worked without that, in fact, I can tell you that we, so we have a nectir channel. Are you familiar with it?

I: Yeah, I've heard about it, I haven't tried it.

UCSB03: It's kind of like Slack, except that it's apparently FERPA compliant. The event, since the term began for this course, We've had 1800 messages from students there.

I: Oh wow.

UCSB03: Yeah, which in some ways is great. So, we have actually three undergraduates who are each spending I don't know something like 10 hours a week, hanging out to answer questions from the students, and that's invaluable. One of the things that's much harder right at the moment is we really really encourage students to work together. You know they have to all turn in their own work but they should certainly be talking to each other. And that's very hard to arrange under current circumstances because they can't physically get together.

I: Yeah, it's something that's hard with our workshops too. So those, the undergraduates that are, you know, supporting nectir is that for just the one course that you're teaching or all of the courses?

UCSB03: That's all for my one course and we're at the end of the second week.

I: That's incredible.

UCSB03: Yeah, yeah, we, I mean, we couldn't do the course without what they're doing for us.

I: Yeah I have had no clue how you know the different departments have been adapting for teaching because we've, unfortunately, had to put so much time into setting up our own remote lab. It takes a lot of time, I believe it took you know, all of those students to do it the whole summer.

UCSB03: So, I've been fortunate because my department chair has just been very supportive of giving us the resources we need to make this work.

I: That's good to hear. I know as since I'm new to the university, It seems like it's a little disjointed between some of the departments. They're run so differently.

UCSB03: Yep. That's a very nice mild way of putting it. Yeah.

UCSB03: Alright, so the next question I have here is how do the ways in which you teach with data relate to the learning goals in your field? Do you teach them to think critically about the sources and uses of data or other specific data skills that prepare them for future careers? I know you touched on R, so anything else?

UCSB03: One of the things in the [course redacted] that is very central is to help students, let me give you the cliché version which is that correlation does not imply causation. And trying to get that through to people is unbelievably hard. And so that's an important learning goal. But for me, an important learning goal in both these courses is that students walk away with tools they can actually use themselves, not just hired of the business and could go run a basic regression and interpret the regression. In this new course, they are learning R. It's kind of a, in principle, we're trying to talk to them about the concepts of how to mess with data. In practice, they're spending a huge amount of their time on R, but I explicitly said there's a final project. And the final project will be a little longer than the homework they've done and one of the goals, is that they should end up with some code and a written document that they could use as part of a job interview. So they can actually show something that they have done.

I: So do you encourage them to put it up on GitHub, so that it's public or to have a place where they can share it from?

UCSB03: No, we don't. We, internally, we're using GitHub, but we've stayed away from it as one too many tech tools to teach.

I: Yeah, I can, I can imagine that would be overloading at that point. Yes, and I, and I meant to ask earlier. Do you have push from, from businesses or anywhere else for using Python or is it mostly R?

UCSB03: Well we've had very little push of any kind, the push we have had was sort of on the level of R. Now the computer science department teaches Python which I think clearly is the right decision. Because that's a very good way to just teach the principles of programming. When I think about the course we're doing, which it's labeled explicitly as we are not doing big data. Okay, I'd like to just learn how to do small data.

I: Which is a much better place to start when you're teaching programming for the first time.

UCSB03: Yeah, I want to say that the actual programming in this course is pretty minimal to somebody who knows how to program. So, it's like the next to last lecture or something that we're going to explain to them what a loop is. And one of the reasons for using R is, to a large extent, you can use it in the same way you would a statistical package, which is, you just have a series of commands that do one thing after another.

I: Yeah I know in our R workshops we're doing one right now, actually I was teaching that earlier today. We rarely, if at all, use loops at all because with R there are so many packages available. You really don't need to know how to do loops and that's one of the I think being big hang ups with programming

UCSB03: I agree, yeah, absolutely.

I: All right. And then kind of a follow up question but slightly different direction. Have you observed any policies or cultural changes, either here at UCSB or in general that influenced the ways that you teach with data?

UCSB03: No, only one, and I'm not quite sure this is responsive, but I do make an attempt to see to it that there are examples that are I guess I'll use the word inclusive. I'll give you an example of what we're doing in this new course. All the students are told that they have to take an implicit bias test which we provide online. And then we're going to take that data and have them analyze it. It's not that the analysis is messy, the analysis could have been done on some other data set. But it seemed like a good opportunity to bring in some other lessons in a very gentle way.

I: Yeah, I think that sounds like a great activity. Because you're, you know you're bringing in the implicit bias and doing that training but you're also analyzing the same data. And, like you said, like having a connection with the student grades, that helps with motivation too.

UCSB03: Yeah, yeah, that's right. Yeah.

I: And then, you know, slightly different question, but along the same lines, do instructors in your field, you know in [discipline redacted] specifically face any ethical challenges with teaching with data? I know we mentioned, you know, anonymization being one of them.

UCSB03: I would say, anonymization. I can't think of really other ethical things that I would call ethical challenges.

I: I guess, do you have, is it often data about individuals?

UCSB03: Yes, very often about individuals. So, in the [discipline redacted], a number of the examples I use data with from what's called the Current Population Survey. You probably know about that. And I often do things like gender wage gaps or things like that. So, I will always say much of the data is actually about individuals but never identified individuals.

I: Okay, never identified ones. So do you end up teaching about anonymization in your courses?

UCSB03: Nope.

I: Alright. So then, I mean the next questions here are on training and support for those courses and I know you mentioned already, the students that you have helping out at least you know with these asynchronous courses. But, do you have any other people that provide instruction or support, you know now or before?

UCSB03: We always have graduate teaching assistants, who play an incredibly important role in meeting with sections, you know, answering student questions, that sort of thing.

I: How big are the classes usually?

UCSB03: The [discipline redacted] is typically 200. And this new class, I think we're down to about 165. Some of the students decided to drop out, which I can understand. Yeah, and so the new one, one of the things that's going to be a challenge going forward is you need to really, I think, need to change the homework each year. Because, as you probably know everything floats around, nowadays. And developing this homework is no longer just a matter of getting an idea. You have to get an idea, get some data, put together. And then, We're using automated grading for the computer code. Which means somebody actually has to write the code that does the coding, the grading code. And so it's going to need on an ongoing basis, some more technical support than most of our courses.

I: Right. It's such a lifesaver to have those though but yeah it does take a lot of work still to keep them up. Do you encourage them to go to drop in sessions around campus or in the [discipline redacted] department, I don't know if there are drop in sessions in the [discipline redacted] department that are available.

UCSB03: Well, in the [discipline redacted] they typically are not, I mean the graduate students and me, will have office hours. And students are certainly, you know, welcome to go to the office hours and they generally don't. In the new class, the original plan before we all got Covid was, we have an undergraduate computer lab. And that was going to be staffed with some undergraduates as many hours as we could manage. So that exactly that would be a place for students to just drop in with questions, well we can't do that. So that's what this nectir channel we're using as a substitute for, and I gotta tell you the undergraduates we've hired to do this, they're fantastic.

I: we have a couple undergraduate student workers too and they're wonderful.

UCSB03: But you know there's no substitute for personal contact. Yeah, it's no great substitute, and, you know, being able to sit down in front of a computer screen and say, here's what I did, I don't know what's going on, what I did wrong. That doesn't work so well online. And we're actually just in fact I think, with some of the undergraduates' advice, we're going to add in some more dropping office hours, where they'll do zoom sessions instead of just this nectir channel. So we're trying hard to keep the students as connected as possible. Once a week, I have an hour, which I have labeled relaxing beverage hour, one student at least has actually taken me up on, where it's like office hours but actually the rule is, we'll talk about anything except for coding and R and the homework. And so it's a way to get to talk to the students about whatever they want to talk about. And so students come in with all sorts of just different questions which is nice.

I: So is that for your undergraduate student workers or the whole class?

UCSB03: No, no, no, no that's for the class.

I: Oh, that's wonderful.

UCSB03: Not that many of them show up, but more show up than used to show up for office hours.

I: Yeah I think everyone's kind of missing the lack of connection and like conversations from the beginning and end of the class are really like are part of the learning process.

UCSB03: They very much are.

I: Yeah, we've had to do a lot of workarounds for these carpentry workshops too, you know, a lot of times such simple questions that people have or like just a spelling error in their code, and unless you can see their screen, you would have no idea how to, how to help them.

UCSB03: Yeah. So one of the things which nearly is I can tell from what my students are explaining to me on how they've arranged the technology. So I don't know if you know UCSB uses a product called Gradescope for grading. And okay, I realize you guys are not grading.

I: No we're not. And I'm pretty new here too.

UCSB03: Yeah. And, but part of what this is, this is a place where students can actually upload their programs. And the idea is the machine will grade it but it also turns out it means that the teaching assistance, and my staff can go in and they can actually while they're trying to answer a student question they can see what the students submitted so they at least have some shot at seeing the code sometimes.

I: Right. And then, do you know if there are other ways that they're learning outside of the formal coursework here? Like for the students that do come in with some knowledge too, do you know of the ways that they're learning or is it self-taught?

UCSB03: I don't know, I mean I will say, one of the other things we did as it was one of the, my students who suggested this something called I think Groupme, where we set up something with the students can actually get together in groups, which we cannot participate in which is good, so they can talk without worrying about us, you know, what we're thinking about them. We are putting up a few links to some other videos online since there are lots on R. It's also clear some of the students have come in with significant backgrounds, some of them know the R coding just fine. But I think most of them don't.

I: Okay. And do you often encourage them to watch those videos? I mean I imagine you do since you put them up

UCSB03: Well, we've been encouraging them, and this is sort of new because again, this is the first time we've done this course and we're sort of learning. One of the things we're learning is that getting started doing this is even harder than we thought it was going to be for the students. So we put a lot of effort into doing hand holding, especially at the beginning. And it turns out, it hasn't been enough. So as we're going we keep trying to add some more hand holding things.

I: Some more materials?

UCSB03: Yes.

I: Yeah, and I know getting started is also a challenge for these workshops we put on and you need to be very explicit and, you know, especially online if you skip a step then then people are lost and you don't even know that they're lost.

UCSB03: Yes.

I: All right, and then as far as your own training goes. Have you received any training in teaching with data or, you know, any more data analysis techniques or software since your graduate degree?

UCSB03: I'm not quite sure how to answer that. I've helped develop some software. I mean, you know, developing techniques for statistical analysis is a half or two thirds of what I do for a living.

I: Okay. So this question is probably not as related to your work since that's, yeah an integral part of it. Yeah. I think I think yeah they meant for this question, you know, other conferences or workshops or, you know, institutions that put on training.

UCSB03: No, I mean I you know, I never programmed in R more than about three lines before teaching this course so you know I went out and bought some books and learned and taught myself.

I: So you mostly use books for that?

UCSB03: Yeah, I, you know I watched some online videos that I thought were useful. You know the folks who do R studio have lots of good resources.

I: Yeah, Rstudio has great resources. And do you end up, do you access them through the UCSB library or get them for yourself?

UCSB03: I just get them for myself. Yeah.

I: I mean we do these workshops, but I need to go through and look more at what we have in the library here. I just haven't quite had time to do that since I've started working here.

I: That's good to hear. Okay, the second to last question, considering the trends in your field, it's less on you know, the training that's available but what should be available. What kinds of training or assistance do you think would be beneficial to you, or other instructors in teaching with data.

UCSB03: It's very clear, it's the tech support sort of not generic but focused for people helping create the course and create all the technical stuff we need for the course. And then the other issue is access to computing facilities for the students. So, you know, we are managing somewhat because Google made this nice donation to us. Otherwise I don't know what we would do. But this of course relies on the students all having good internet access and having a laptop or something anyhow. And so, that second part could be a bit of an issue but it's really the people costs that I'm concerned about. Yeah.

I: Yeah, I know. You know, we've set up this remote lab because we also got a gift from Microsoft Azure, very timely for all of this for remote computing but I'm surprised at how little there was for remote computing, when I got the campus here.

UCSB03: That's very true. And they may, I mean how far they've come in six months is amazing.

I: Right. It's really prioritized. Yeah, set the priorities. But the manpower is a hard part of it for sure, and you know it doesn't help if like you said, the students don't have access to the internet or a computer that can connect.

UCSB03: Right. And you know, let's assume that next year, this will all be over and we'll be back in person which will be great, but then we're back at the issue of computer labs and things like that, where students can get together and where they can actually get help. So for example the open access labs on campus are generally only open Monday through Friday. Right. And you know I understand this is a cost reason for that, that's not criticizing, and they're also open mostly just during the day. Well, it would be a big help if there was just a lot more accessibility to computers and stuff like that for the undergraduates.

I: Yeah, and I know on my part, it seems like the these open computer labs that are available on campus are, you know, so all over the place and decentralized that it took me forever to get a grasp of where there were open computing labs and you know I'm still finding out about different labs that are you know were available on campus before we went remote. And I don't know if the students have an easier way of finding that out then I do as staff but...

UCSB03: I mean, the letters and sciences actually runs a whole bunch of the labs and have a, well not a very good website telling you where they are when they're open and what they have. And then of course different departments have their own labs that are just for internal department use.

I: So actually your undergrad lab and [discipline redacted] how many computers, do you have there?

UCSB03: I'm not sure I think they're 35.

I: Okay.

UCSB03: And again, a lot of this was we have some donors who have given us some money to do some of it so you know we're fortunate in that way.

I: That always helps. Yeah. Well, those I mean those are all the questions I have and the final thing is just, is there anything else from your experiences and perspectives that you can think of that might be relevant to this or something that you've thought of from a question earlier that you'd like to share?

UCSB03: No, no, I, I think you've covered most of the stuff. I guess I should say, by way of background. That [professional redacted] typically do not do huge amounts of original data collection.

I: Okay.

UCSB03: We tend to use data that somebody else's collected. There are exceptions to that. My daughter is an [professional redacted] and runs a survey crew in [country redacted]. But that's why some of my answers might be different from some of the other social science groups and anonymization just doesn't come up all that much compared to places where you're well, compared to like what you're doing right now.

I: That's really interesting because yeah I'm not familiar with the ins and outs of, you know, [discipline redacted] data or, you know, analysis versus other departments, it's just it's you know it's not what I studied and, and we have a lot of the [discipline redacted] related software in the Collaboratory so it's hard because you know we have the software but I don't have the knowledge to know how it might be used.